WASP-12b

R-1613

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_190203` · created 2026-08-02T03:40:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458517.76195 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.043
Transit depth (Rp/Rs)^2	1.57 +/- 1.08 [%]
Orbital Inclination (inc)	86.46 +/- 2.95
Airmass coefficient 1 (a1)	1.032 +/- 0.046
Airmass coefficient 2 (a2)	-0.031 +/- 0.033
Scatter in the residuals of the lightcurve fit is	4.51 %
Best Comparison Star	#1 - [330, 224]
Optimal Aperture	1.94
Optimal Annulus	10.8
Transit Duration (day)	0.129 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.043 vs 0.1178 (deviation 0.16σ)
Duration fit vs published	0.129 d vs 0.125 d (deviation 0.62σ)
Mid-time O–C	0.9 min
Depth SNR	2.8
Residual scatter	4.51 %

Light curve

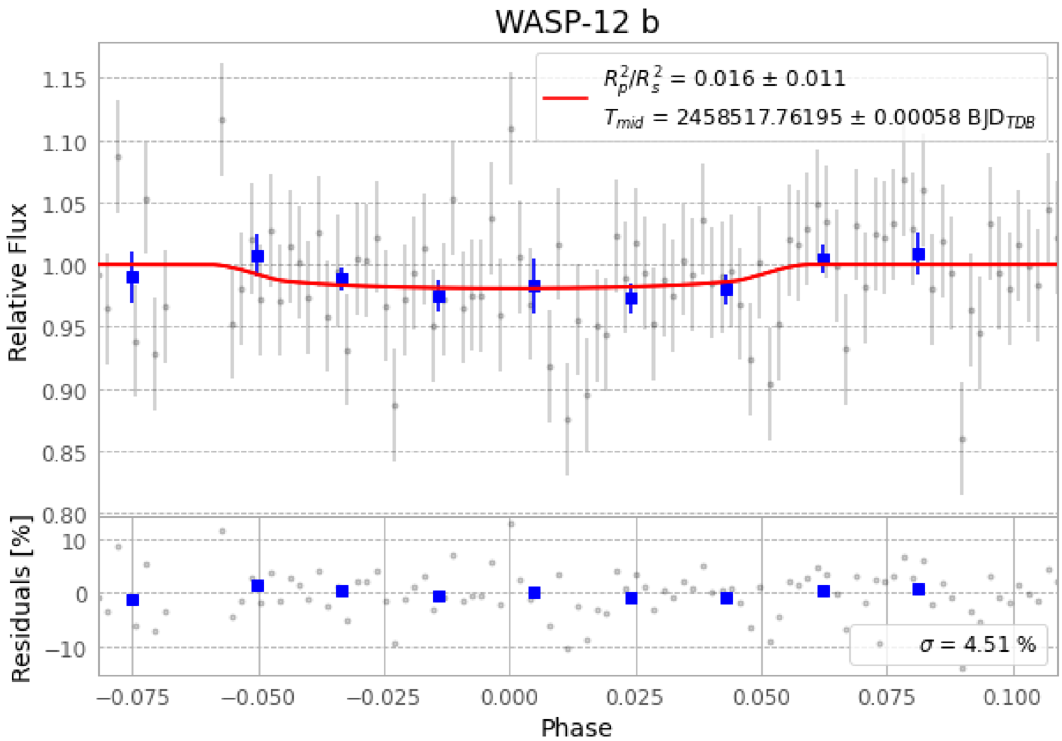

Figure 1 — FinalLightCurve_WASP-12 b_2019-02-02.png (SHA-256 e4326535b4faac91…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 187], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e37a529ada59427a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

101 FITS frames (66 MB), `WASP-12190203040012.FITS` → `WASP-12190203090013.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190203.log (2f0ed1fa9ce3…), FinalLightCurve_WASP-12 b_2019-02-02.png (e4326535b4fa…), FinalLightCurve_WASP-12 b_2019-02-02.csv (f7c378273476…)

Compute resources

Wall time 360.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)